ECE30030/ITP30010 Database Systems

SQL DML

Reading: Chapter 3

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Spring, 2026

Handong Global University

Announcements

- TA office hours are available on a regular basis
  - Monday 14:00-15:00 -- Eunchan Jung (NTH217; 22100685@handong.ac.kr)
  - Tuesday 19:00-20:00 -- Hyunmo Kang (NTH217; 22100026@handong.ac.kr)
  - Thursday 14:00-15:00 -- Yunseong Choi (NTH221; yunseong.choi@handong.ac.kr)
  - Friday 14:00-15:00 -- Yujin Kim (NTH221; yujin.kim@handong.ac.kr)

  - If you find any part of the material challenging or would like additional help, please feel encouraged to visit

Announcements

- Term project will kick off soon
  - The project will be conducted in teams of three students
    - Teams may be formed by students, assigned by the instructor, or partially arranged
    - If you have preferred teammates, please discuss and coordinate in advance
  - The project will run throughout the semester
  - It consists of two phases (with two submissions and evaluations)
    - Phase 1: Designing a database schema
    - Phase 2: Optimizing query performance
  - At the end of the semester, self and peer evaluations will be conducted and reflected in individual grades

Agenda

- SQL data manipulation language (DML)
  - Aggregate functions, aggregation
  - Insert, update, delete
- Nested subqueries
- Set membership (SOME, ALL, EXISTS)

Running Examples

- Relations (tables): *instructor*, *teaches*

Instructor relation

ID	name	dept_name	salary
10101	Srinivasan	Comp. Sci.	65000.00
12121	Wu	Finance	90000.00
15151	Mozart	Music	40000.00
22222	Einstein	Physics	95000.00
32343	El Said	History	60000.00
33456	Gold	Physics	87000.00
45565	Katz	Comp. Sci.	75000.00
58583	Califieri	History	62000.00
76543	Singh	Finance	80000.00
76766	Crick	Biology	72000.00
83821	Brandt	Comp. Sci.	92000.00
98345	Kim	Elec. Eng.	80000.00

teaches relation

ID	course_id	sec_id	semester	year
76766	BIO-101	1	Summer	2017
76766	BIO-301	1	Summer	2018
10101	CS-101	1	Fall	2017
45565	CS-101	1	Spring	2018
83821	CS-190	1	Spring	2017
83821	CS-190	2	Spring	2017
10101	CS-315	1	Spring	2018
45565	CS-319	1	Spring	2018
83821	CS-319	2	Spring	2018
10101	CS-347	1	Fall	2017
98345	EE-181	1	Spring	2017
12121	FIN-201	1	Spring	2018
32343	HIS-351	1	Spring	2018
15151	MU-199	1	Spring	2018
22222	PHY-101	1	Fall	2017

Running Examples

- Relations (tables): *course*, *takes*

course relation

course_id	title	dept_name	credits
BIO-101	Intro. to Biology	Biology	4
BIO-301	Genetics	Biology	4
BIO-399	Computational Biology	Biology	3
CS-101	Intro. to Computer Science	Comp. Sci.	4
CS-190	Game Design	Comp. Sci.	4
CS-315	Robotics	Comp. Sci.	3
CS-319	Image Processing	Comp. Sci.	3
CS-347	Database System Concepts	Comp. Sci.	3
EE-181	Intro. to Digital Systems	Elec. Eng.	3
FIN-201	Investment Banking	Finance	3
HIS-351	World History	History	3
MU-199	Music Video Production	Music	3
PHY-101	Physical Principles	Physics	4

takes relation

ID	course_id	sec_id	semester	year	grade
00128	CS-101	1	Fall	2017	A
00128	CS-347	1	Fall	2017	A-
12345	CS-101	1	Fall	2017	C
12345	CS-190	2	Spring	2017	A
12345	CS-315	1	Spring	2018	A
12345	CS-347	1	Fall	2017	A
19991	HIS-351	1	Spring	2018	B
23121	FIN-201	1	Spring	2018	C+
44553	PHY-101	1	Fall	2017	B-
45678	CS-101	1	Fall	2017	F
45678	CS-101	1	Spring	2018	B+
45678	CS-319	1	Spring	2018	B
54321	CS-101	1	Fall	2017	A-
54321	CS-190	2	Spring	2017	B+
55739	MU-199	1	Spring	2018	A-
76543	CS-101	1	Fall	2017	A
76543	CS-319	2	Spring	2018	A
76653	EE-181	1	Spring	2017	C
98765	CS-101	1	Fall	2017	C-
98765	CS-315	1	Spring	2018	B
98988	BIO-101	1	Summer	2017	A
98988	BIO-301	1	Summer	2018	<null>

Running Examples

- Relations (tables): *student*

student relation

ID	name	dept_name	tot_cred
00128	Zhang	Comp. Sci.	102
12345	Shankar	Comp. Sci.	32
19991	Brandt	History	80
23121	Chavez	Finance	110
44553	Peltier	Physics	56
45678	Levy	Physics	46
54321	Williams	Comp. Sci.	54
55739	Sanchez	Music	38
70557	Snow	Physics	0
76543	Brown	Comp. Sci.	58
76653	Aoi	Elec. Eng.	60
98765	Bourikas	Elec. Eng.	98
98988	Tanaka	Biology	120

Aggregate Functions

- These functions operate on the multiset of values of a column of a relation, and return a value
  - **AVG:** average value
  - **MIN:** minimum value
  - **MAX:** maximum value
  - **SUM:** sum of values
  - **COUNT:** number of values

Aggregate Functions Examples

- Find the average salary of instructors in the Computer Science department
  - **SELECT AVG**(*salary*)
    FROM *instructor*
    WHERE *dept_name*= 'Comp. Sci.';

`AVG(salary)`
77333.333333

- Find the total number of instructors who teach a course in the Spring 2018 semester
  - **SELECT COUNT**(**DISTINCT** *ID*)
    FROM *teaches*
    WHERE *semester* = 'Spring' **AND** *year* = 2018;

`COUNT(DISTINCT ID)`
6

- Find the number of tuples in the *teaches* relation
  - **SELECT COUNT** (*)
    FROM *teaches*;

`COUNT(*)`
15

Aggregate Functions: GROUP BY

- Find the average salary of instructors in each department
  - **SELECT** *dept_name*, **AVG**(*salary*) **AS** *avg_salary*
    FROM *instructor*
    GROUP BY *dept_name*;

ID	*name*	*dept_name*	*salary*
76766	Crick	Biology	72000
45565	Katz	Comp. Sci.	75000
10101	Srinivasan	Comp. Sci.	65000
83821	Brandt	Comp. Sci.	92000
98345	Kim	Elec. Eng.	80000
12121	Wu	Finance	90000
76543	Singh	Finance	80000
32343	El Said	History	60000
58583	Califieri	History	62000
15151	Mozart	Music	40000
33456	Gold	Physics	87000
22222	Einstein	Physics	95000

dept_name	avg_salary
Biology	72000.000000
Comp. Sci.	77333.333333
Elec. Eng.	80000.000000
Finance	85000.000000
History	61000.000000
Music	40000.000000
Physics	91000.000000

Aggregation

- Attributes in **SELECT** clause outside of aggregate functions must appear in **GROUP BY** list
  - /* erroneous query */
    SELECT *dept_name*, *ID*, **AVG**(*salary*)
    FROM *instructor*
    GROUP BY *dept_name*;

dept_name	ID	`AVG(salary)`
Biology	76766	72000.000000
Comp. Sci.	10101	77333.333333
Elec. Eng.	98345	80000.000000
Finance	12121	85000.000000
History	32343	61000.000000
Music	15151	40000.000000
Physics	22222	91000.000000

Aggregate Functions – Having Clause

- Find the names and average salaries of all departments whose average salary is greater than 65000
  - **SELECT** *dept_name*, **AVG**(*salary*) **AS** *avg_salary*
    FROM *instructor*
    GROUP BY *dept_name*
    HAVING AVG(*salary*) > 65000;

dept_name	avg_salary
Biology	72000.000000
Comp. Sci.	77333.333333
Elec. Eng.	80000.000000
Finance	85000.000000
Physics	91000.000000

Aggregate Functions – Having Clause

- Note: predicates in the **HAVING** clause are applied after the formation of groups whereas predicates in the **WHERE** clause are applied before forming groups

SELECT *dept_name*, **AVG**(*salary*) **AS** *avg_salary*
FROM *instructor*
GROUP BY *dept_name*
HAVING AVG(*salary*) > 65000;

dept_name	avg_salary
Biology	72000.000000
Comp. Sci.	77333.333333
Elec. Eng.	80000.000000
Finance	85000.000000
Physics	91000.000000

SELECT *dept_name*, **AVG**(*salary*) **AS** *avg_salary*
FROM *instructor*
WHERE *salary* > 65000
GROUP BY *dept_name*;

dept_name	avg_salary
Biology	72000.000000
Comp. Sci.	83500.000000
Elec. Eng.	80000.000000
Finance	85000.000000
Physics	91000.000000

Agenda

- SQL data manipulation language (DML)
  - Aggregate functions, aggregation
  - **Insert, update, delete**
- Nested subqueries
- Set membership (SOME, ALL, EXISTS)

INSERT

- Basic syntax
  - Insert data into every column:
    - **INSERT INTO** *tablename*
      VALUES (*col1_value, col2_value, …*)
    - Must list values in the same order as in the table schema
    - If some data values are unknown, must type NULL
    - For character sequences, use quotation marks
      - Single quotation marks are preferred (but double quotation marks are allowed)
      - Value in quotations is case-sensitive
  - Insert data into selected columns
    - **INSERT INTO** *tablename* (*col1_name, col3_name, col4_name, …*)
      VALUES (*col1_value, col3_value, col4_value, …*)

INSERT

- Add a new tuple to *course*
  - **INSERT INTO** *course*
    VALUES ('CS-437', 'Database Systems', 'Comp. Sci.', 4);

- or equivalently
  - **INSERT INTO** *course* (*course_id*, *title*, *dept_name*, *credits*)
    VALUES ('CS-437', 'Database Systems', 'Comp. Sci.', 4);

- Add a new tuple to *student* with *tot_creds* set to null
  - **INSERT INTO** *student*
    VALUES ('3003', 'Green', 'Finance', *null*);

INSERT

- A foreign key specifies that an attribute from one relation has to map to a tuple in another relation
  - Value in one relation must appear in another relation
    - Referencing relation → Referenced relation

Relation: instructor

ID	name ▲ 2	dept_name ▲ 1	salary
76766	Crick	Biology	72000.00
83821	Brandt	Comp. Sci.	92000.00
45565	Katz	Comp. Sci.	75000.00

Relation: department

dept_name	building	budget
Biology	Watson	90000.00
Comp. Sci.	Taylor	100000.00
Elec. Eng.	Taylor	85000.00

INSERT

- A foreign key specifies that an attribute from one relation has to map to a tuple in another relation
  - Value in one relation must appear in another relation
    - Referencing relation → Referenced relation

- Make sure all foreign keys that new row references have already been added to database
  - One cannot insert a foreign key value unless the corresponding value exists in the referenced relation

C.f., Primary Keys

- A relation's primary key uniquely identifies a single tuple
- Some DBMSs automatically create an internal primary key if you do not define one
  - *E.g.*, SQL:2003 (SEQUENCE), MySQL (AUTO_INCREMENT)

- Example
  - instructor(ID, name, dept_name, salary)

ID	name 2	dept_name 1	salary
76766	Crick	Biology	72000.00
83821	Brandt	Comp. Sci.	92000.00
45565	Katz	Comp. Sci.	75000.00

INSERT

- Inserting results of other SELECT query
  - Make each student in the Music department who has earned more than 144 credit hours an instructor in the Music department with a salary of $18,000
    - **INSERT INTO** *instructor*
      SELECT *ID, name, dept_name, 18000*
      FROM *student*
      WHERE *dept_name* = 'Music' **AND** *total_cred* > 144;
  - The **SELECT FROM WHERE** statement is evaluated fully before any of its results are inserted into the relation
    - Otherwise queries like
      INSERT INTO *table*1 **SELECT** * **FROM** *table*1
      would cause problem

UPDATE

- Basic syntax
  - Updating a table
    - **UPDATE** *tablename*
      SET *col1_name = new_col1_value, col2_name = new_col2_value, …*;
  - Updating a table with conditions
    - **UPDATE** *tablename*
      SET *col1_name = new_col1_value, col2_name = new_col2_value, …*
      WHERE *predicate*;

UPDATE

- Give a 5% salary raise to all instructors
  - **UPDATE** *instructor*
    SET *salary* = *salary* * 1.05

- Give a 5% salary raise to those instructors who earn less than 70000
  - **UPDATE** *instructor*
    SET *salary* = *salary* * 1.05
    WHERE *salary* < 70000;

- Give a 5% salary raise to instructors whose salary is less than average
  - **UPDATE** *instructor*
    SET *salary* = *salary* * 1.05
    WHERE *salary* < (**SELECT AVG**(salary) **FROM** *instructor*);

UPDATE

- Increase salaries of instructors whose salary is over $100,000 by 3%, and all others by a 5%
  - Write two UPDATE statements:

```
UPDATE instructor
SET salary = salary * 1.03
WHERE salary > 100000;

UPDATE instructor
SET salary = salary * 1.05
WHERE salary <= 100000;
```

  - The order is important
  - Can be done better using the **case** statement (next slide)

CASE Statement for Conditional Update

- The following query is equivalent to the previous UPDATE queries
  - **UPDATE** *instructor*
    SET *salary* = **CASE**
    WHEN *salary* <= 100000 **THEN** *salary* * 1.05
    ELSE *salary* * 1.03
    END

UPDATE with Scalar Subqueries

- Recompute and update *tot_creds* value for all students
  - **UPDATE** *student S*
    SET *tot_cred* = (**SELECT SUM**(*credits*)
    FROM *takes, course*
    WHERE *takes.course_id* = *course.course_id* **AND**
    S.ID= *takes.ID* **AND**
    takes.grade <> 'F' **AND**
    takes.grade **IS NOT NULL**);

DELETE

- Basic syntax
  - To remove specific rows
    - **DELETE FROM** *tablename*
      WHERE *predicate*;
  - To remove all rows
    - **DELETE FROM** *tablename*;
    - This is equivalent to **TRUNCATE**:
      TRUNCATE (**TABLE**) *tablename*;
    - One cannot truncate a table with foreign key constraints
      - Must disable the constraints first (we will cover **ALTER** when we study SQL DDL):
        ALTER TABLE *tablename*
        DISABLE CONSTRAINT *constraint_name*;

DELETE

- Delete all instructors
  - **DELETE FROM** *instructor*;

- Delete all instructors from the Finance department
  - **DELETE FROM** *instructor*
    WHERE *dept_name*= 'Finance';

- Delete all tuples in the instructor relation for those instructors associated with a department located in the Watson building
  - **DELETE FROM** *instructor*
    WHERE *dept name* **IN** (**SELECT** *dept name*
    FROM *department*
    WHERE *building* = 'Watson');

DELETE

- Delete all instructors whose salary is less than the average salary of instructors
  - Example: **DELETE FROM** *instructor*
    WHERE *salary* < (**SELECT AVG** (*salary*)
    FROM *instructor*);

- Issue: as we delete tuples from *instructor*, the average salary changes
  - Solution used in SQL:
    1. First, compute **AVG**(*salary*) and find all tuples to delete
    2. Next, delete all tuples found above (without recomputing **AVG** or retesting the tuples)

Agenda

- SQL data manipulation language (DML)
  - Aggregate functions, aggregation
  - Insert, update, delete
- **Nested subqueries**
- Set membership (SOME, ALL, EXISTS)

Nested Subqueries

- SQL provides a mechanism for the nesting of subqueries. A subquery is a SELECT-FROM-WHERE expression that is nested within another query
- The nesting can be done in the following SQL query

  SELECT $A_1, A_2, \ldots, A_n$
  FROM $r_1, r_2, \ldots, r_m$
  WHERE P

  as follows:
  - **FROM clause:** r_i can be replaced by any valid subquery
  - **WHERE clause:** P can be replaced with an expression of the form:
    B <operation> (subquery)
    B is an attribute and <operation> is to be defined later
  - **SELECT clause:**
    A_i can be replaced by a subquery that generates a single value (scalar subquery)

Subqueries in the FROM Clause

- Find the average instructors' salaries of those departments where the average salary is greater than $42,000
  - **SELECT** *D.dept_name*, *D.avg_salary*
    FROM (**SELECT** *dept_name*, **AVG**(*salary*) **AS** *avg_salary*
    FROM *instructor*
    GROUP BY *dept_name*) **AS** *D*
    WHERE *D.avg_salary* > 42000;

dept_name	avg_salary
Biology	72000.000000
Comp. Sci.	77333.333333
Elec. Eng.	80000.000000
Finance	85000.000000
History	61000.000000
Physics	91000.000000

Scalar Subquery

- Scalar subquery is used where a single value is expected
  - Runtime error occurs if a subquery returns more than one result tuple
- List all departments along with the number of instructors in each department
  - **SELECT** *dept_name*,
    (**SELECT COUNT**(*)
    FROM *instructor*
    WHERE *department.dept_name* = *instructor.dept_name*)
    AS *num_instructors*
    FROM *department*;

dept_name	num_instructors
Biology	1
Comp. Sci.	3
Elec. Eng.	1
Finance	2
History	2
Music	1
Physics	2

WITH Clause

- The **WITH** clause provides a way of defining a temporary relation
  - The relation is available only to the query in which the **WITH** clause occurs

- Find all departments with the maximum budget
  - **WITH** *max_budget* (*value*) **AS**
    (**SELECT MAX**(*budget*)
    FROM *department*)
    SELECT *department.dept_name*
    FROM *department*, *max_budget*
    WHERE *department.budget* = *max_budget.value*;

dept_name
Finance

Common Table Expression (CTE)

- Common Table Expression (CTE): A temporary named result set
  - Exists only during the execution of a single query
  - Can be referenced multiple times within the query
  - Computed only during query execution
    - CTE is not stored (unlike a view)
    - Supports recursive queries

  - **WITH** temp **AS** (...)
    SELECT * **FROM** temp;

- Why it is useful
  - CTEs improve query structure by:
    - Making complex queries more readable
    - Breaking logic into step-by-step components
    - Avoiding repeated subqueries

Agenda

- SQL data manipulation language (DML)
  - Aggregate functions, aggregation
  - Insert, update, delete
- Nested subqueries
- **Set membership (SOME, ALL, EXISTS)**

Set Membership

- Find courses offered in Fall 2017 and in Spring 2018
  - **SELECT DISTINCT** *course_id*
    FROM *teaches*
    WHERE *semester* = 'Fall' **AND** *year*= 2017 **AND**
    course_id **IN** (**SELECT** *course_id*
    FROM *teaches*
    WHERE *semester* = 'Spring' **AND** *year*= 2018);

course_id
CS-101

Set Membership

- Find courses offered in Fall 2017 but not in Spring 2018
  - **SELECT DISTINCT** *course_id*
    FROM *teaches*
    WHERE *semester* = 'Fall' **AND** *year*= 2017 **AND**
    course_id ***NOT IN*** (**SELECT** *course_id*
    FROM *teaches*
    WHERE *semester* = 'Spring' **AND** *year*= 2018);

course_id
CS-347
PHY-101

Set Membership

- Name all instructors whose name is neither “Mozart” nor Einstein”
  - **SELECT DISTINCT** *name*
    FROM *instructor*
    WHERE *name* **NOT IN** ('Mozart', 'Einstein’);

name
Srinivasan
Wu
El Said
Gold
Katz
Califieri
Singh
Crick
Brandt
Kim

name
Srinivasan
Wu
Mozart
Einstein
El Said
Gold
Katz
Califieri
Singh
Crick
Brandt
Kim

Set Membership

- Find the total number of unique students who have taken course sections taught by the instructor with *ID* 10101
  - **SELECT COUNT**(**DISTINCT** *ID*)
    FROM *takes*
    WHERE (*course_id*, *sec_id*, *semester*, *year*) **IN**
    (**SELECT** *course_id*, *sec_id*, *semester*, *year*
    FROM *teaches*
    WHERE *teaches*.*ID*= 10101);

`COUNT(DISTINCT ID)`
6

  - Note: *Above query can be written in a much simpler manner*
    The formulation above is simply to illustrate SQL features

Set Comparison – SOME

- Find names of instructors with salary greater than that of SOME (at least one) instructor in the Biology department
  - **SELECT DISTINCT** *T.name*
    FROM *instructor* **AS** *T*, *instructor* **AS** *S*
    WHERE *T.salary* > *S.salary* **AND** *S.dept name* = 'Biology';

- Same query using > **SOME** clause
  - **SELECT** *name*
    FROM *instructor*
    WHERE *salary* > **SOME** (**SELECT** *salary*
    FROM *instructor*
    WHERE *dept_name* = 'Biology');

name
Wu
Einstein
Gold
Katz
Singh
Brandt
Kim

Interpretation of SOME

- F <comp> **SOME** $r \Leftrightarrow \exists t \in r$ such that (F <comp> t)
  Where <comp> can be: $<, \leq, >, =, \neq$

(5 < **SOME** $\boxed{0 \mid 5 \mid 6}$) = true (read: 5 < some tuple in the relation)

(5 < **SOME** $\boxed{0 \mid 5}$) = false

(5 = **SOME** $\boxed{0 \mid 5}$) = true

(5 ≠ **SOME** $\boxed{0 \mid 5}$) = true (since 0 ≠ 5)

(= **SOME**) ≡ **IN**
However, (≠ **SOME**) ≢ **NOT IN**

Set Comparison – ALL

- Find the names of ALL instructors whose salary is greater than the salary of ALL instructors in the Biology department
  - **SELECT** *name*
    FROM *instructor*
    WHERE *salary* > **ALL** (**SELECT** *salary*
    FROM *instructor*
    WHERE *dept name* = 'Biology');

name
Wu
Einstein
Gold
Katz
Singh
Brandt
Kim

Interpretation of ALL

- F <comp> **ALL** $r \Leftrightarrow \forall t \in r$ (F <comp> t)

(5 < **ALL** $\begin{array}{|c|}\hline 0 \\ \hline 5 \\ \hline 6 \\ \hline\end{array}$) = false

(5 < **ALL** $\begin{array}{|c|}\hline 6 \\ \hline 10 \\ \hline\end{array}$) = true

(5 = **ALL** $\begin{array}{|c|}\hline 4 \\ \hline 5 \\ \hline\end{array}$) = false

(5 ≠ **ALL** $\begin{array}{|c|}\hline 4 \\ \hline 6 \\ \hline\end{array}$) = true (since 5 ≠ 4 and 5 ≠ 6)

(≠ **ALL**) ≡ **NOT IN**
However, (= **ALL**) ≢ **IN**

Test for Empty Relations

- The **EXISTS** construct returns the value *true* if the argument subquery is nonempty
  - **EXISTS** $r \Leftrightarrow r \neq \emptyset$
  - **NOT EXISTS** $r \Leftrightarrow r = \emptyset$

Use of EXISTS

- Yet another way of specifying the query "Find all courses taught in both the Fall 2017 semester and in the Spring 2018 semester"
  - **SELECT** *course_id*
    FROM *teaches* **AS** *S*
    WHERE *semester* = 'Fall' **AND** *year* = 2017 **AND**
    EXISTS (**SELECT** *
    FROM *teaches* **AS** *T*
    WHERE *semester* = 'Spring' **AND** *year*= 2018
    AND *S.course_id* = *T.course_id*);

course_id
CS-101

Use of NOT EXISTS

- Find all students who have taken all courses offered in the Music department
  - **SELECT DISTINCT** *S.ID*, *S.name*
    FROM *student* **AS** *S*
    WHERE NOT EXISTS (**SELECT** *course_id*
    FROM *course*
    WHERE *dept_name* = 'Music'
    AND *course_id* **NOT IN**
    (**SELECT** *T.course_id*
    FROM *takes* **AS** *T*
    WHERE *S.ID* = *T.ID*));

ID	name
55739	Sanchez

Use of NOT EXISTS

- Note: Renaming (**AS**) is optional in certain contexts
  - **SELECT DISTINCT** *ID*, *name*
    FROM *student*
    WHERE NOT EXISTS (**SELECT** *course_id*
    FROM *course*
    WHERE *dept_name* = ‘Music'
    AND *course_id* **NOT IN**
    (**SELECT** *course_id*
    FROM *takes*
    WHERE *student*.*ID* = *takes*.*ID*));

  - Exception: the following query results in an empty relation
    - **SELECT DISTINCT** *name*
      FROM *instructor*
      WHERE *salary* > *salary* **AND** *dept_name* = 'Biology';

Use of NOT EXISTS

- Some systems support the **EXCEPT** clause (MySQL does not)
- Find all students who have taken all courses offered in the Music department
  - **SELECT DISTINCT** *S.ID*, *S.name*
    FROM *student* **AS** *S*
    WHERE NOT EXISTS ((**SELECT** *course_id*
    FROM *course*
    WHERE *dept_name* = ‘Music')
    EXCEPT
    (**SELECT** *T.course_id*
    FROM *takes* **AS** *T*
    WHERE *S.ID* = *T.ID*));

Test for Absence of Duplicate Tuples

- The **UNIQUE** construct tests whether a subquery has any duplicate tuples in its result
  - **UNIQUE** evaluates to "true" if a given subquery contains no duplicates
  - MySQL does not support the UNIQUE test (UNIQUE in MySQL is a constraint specifier)

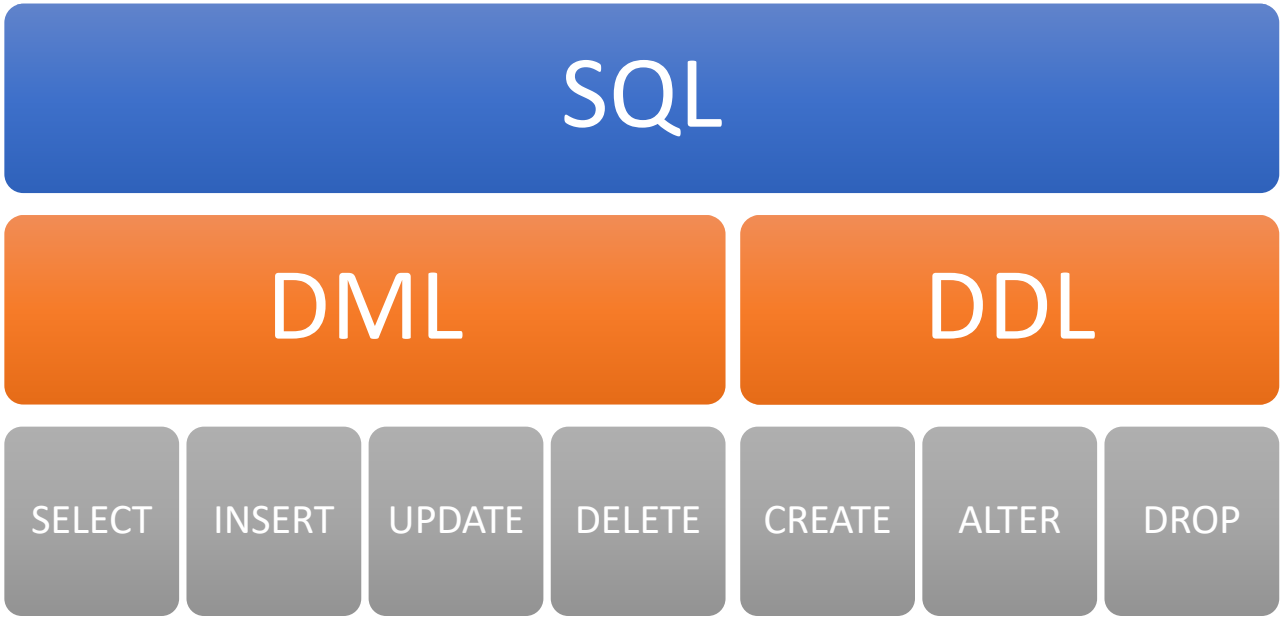
- Find all courses that were offered at most once in 2017
  - **SELECT** *T.course_id*
    FROM *course* **AS** *T*
    WHERE UNIQUE (**SELECT** *R.course_id*
    FROM *teaches* **AS** *R*
    WHERE *T.course_id*= *R.course_id* **AND** *R.year* = 2017);

SQL Order of Execution

Order	Clause	Function
1	FROM	Choose and join tables to get base data
2	WHERE	Filters the base data
3	GROUP BY	Aggregates the base data
4	HAVING	Filters the aggregated data
5	SELECT	Returns the final data
6	ORDER BY	Sorts the final data
7	LIMIT	Limits the returned data to a row count

```
SELECT developer, COUNT(*) AS GamesCount, AVG(Price) AS AvgPrice
FROM games
JOIN distributors ON games.dist_id = distributors.dist_id
WHERE release_year > 2019
GROUP BY developer
HAVING COUNT(*) > 2
ORDER BY AvgPrice DESC;
```

SQL Order of Execution

- Explanation
  1. **FROM** games: The data source is defined
  2. **JOIN** distributors **ON** games.dist_id = distributors.dist_id: Tables are joined to include distributor data in the results
  3. **WHERE** release_year > 2019: Rows are filtered based on the release year of the games
  4. **GROUP BY** developer: The results are grouped by the developer
  5. **HAVING COUNT**(*) > 2: The groups are filtered based on the count of games per developer
  6. **SELECT** developer, **COUNT**(*) AS GamesCount, **AVG**(Price) **AS** AvgPrice: Specific columns and calculated columns are selected
  7. **ORDER BY** AvgPrice **DESC**: The final result set is ordered by the average price in descending order

SQL Commands